

3-node Cassandra cluster before doing the consistency experiments:

```
salma-rabie@salma-rabie-VirtualBox:~/Lab3$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
e6b021a1e325   cassandra:4.1  "docker-entrypoint.s..." 5 minutes ago  Up 5 minutes  7000
-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9040->9042/tcp, [::]:9040->9042/tcp  cassandra2
ce2fa6d9ac00   cassandra:4.1  "docker-entrypoint.s..." 7 minutes ago  Up 7 minutes  7000
-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9041->9042/tcp, [::]:9041->9042/tcp  cassandra1
f22c62757821   cassandra:4.1  "docker-entrypoint.s..." 7 minutes ago  Up 7 minutes  7000
-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9042->9042/tcp, [::]:9042->9042/tcp  cassandra-seed
salma-rabie@salma-rabie-VirtualBox:~/Lab3$ docker exec cassandra-seed nodetool status
Datacenter: se1
=====
Status=Up/Down
|/ State=Normal/Leaving/Joining/Moving
-- Address      Load          Tokens   Owns (effective)  Host ID
Rack
UN  172.18.0.4    70.23 KiB     16       76.0%             f32e8542-89b7-4297-af23-d3ba61b3bcd6
rack1
UN  172.18.0.3    70.22 KiB     16       59.3%             0b1b1cbd-57b0-4fb8-9fd6-cd95bdec4e85
rack1
UN  172.18.0.2    125.43 KiB    16       64.7%             60ed1d18-dc66-4a5d-863e-35ddee56228e
rack1
```

A) RF=1 setup:

```
salma-rabie@salma-rabie-VirtualBox:~/Lab3$ docker exec -it cassandra-seed cqlsh

ATTENTION: All commands will be saved to history file: /root/.cassandra/cqlsh_history
This may include sensitive information such as passwords.
To disable history, use --disable-history or set 'disabled = true' in the [history] section of cqlshrc.
See https://cassandra.apache.org/doc/latest/tools/cqlsh.html for more information.

Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.1.0 | Cassandra 4.1.12 | CQL spec 3.4.6 | Native protocol v5]
Use HELP for help.
cqlsh> CREATE KEYSPACE bookstore
... WITH replication = {
...   'class': 'SimpleStrategy',
...   'replication_factor': 1
... };
cqlsh> USE bookstore;
cqlsh:bookstore> CREATE TABLE books (
...   name VARCHAR,
...   category VARCHAR,
...   year INT,
...   title VARCHAR,
...   PRIMARY KEY ((name), category, year, title)
... );
cqlsh:bookstore> INSERT INTO books (name, category, year, title)
... VALUES ('salma_rf1', 'fiction', 2026, 'Book_RF1');
cqlsh:bookstore> SELECT * FROM bookstore.books
... WHERE name='salma_rf1';

name      | category | year | title
-----+-----+-----+-----
salma_rf1 | fiction  | 2026 | Book_RF1

(1 rows)
cqlsh:bookstore>
```

Set the read consistency to QUORUM:

```
cqlsh:bookstore> CONSISTENCY QUORUM;
Consistency level set to QUORUM.
cqlsh:bookstore> SELECT * FROM bookstore.books
... WHERE name='salma_rf1';

name      | category | year | title
-----+-----+-----+-----
salma_rf1 | fiction  | 2026 | Book_RF1

(1 rows)
cqlsh:bookstore>
```

1- First experiment: Node 1 down

Pause Node 1:

```
salma-rabie@salma-rabie-VirtualBox:~/Lab3$ docker pause cassandra-seed
cassandra-seed
salma-rabie@salma-rabie-VirtualBox:~/Lab3$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        NAMES
PORTS
e6b021a1e325   cassandra:4.1  "docker-entrypoint.s..." 23 minutes ago Up 23 minutes  cassandra2
7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9040->9042/tcp, [::]:9040->9042/tcp
ce2fa6d9ac00   cassandra:4.1  "docker-entrypoint.s..." 25 minutes ago Up 25 minutes  cassandra1
7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9041->9042/tcp, [::]:9041->9042/tcp
f22c62757821   cassandra:4.1  "docker-entrypoint.s..." 25 minutes ago Up 25 minutes (Paused) cassandra-seed
7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9042->9042/tcp, [::]:9042->9042/tcp
salma-rabie@salma-rabie-VirtualBox:~/Lab3$
```

Connect to Node 2:

➤ Consistency level = QUORUM

```
salma-rabie@salma-rabie-VirtualBox:~/Lab3$ docker exec -it cassandra1 cqlsh

ATTENTION: All commands will be saved to history file: /root/.cassandra/cqlsh_history
This may include sensitive information such as passwords.
To disable history, use --disable-history or set 'disabled = true' in the [history] section of cqlshrc.
See https://cassandra.apache.org/doc/latest/tools/cqlsh.html for more information.

Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.1.0 | Cassandra 4.1.12 | CQL spec 3.4.6 | Native protocol v5]
Use HELP for help.
cqlsh> USE bookstore;
cqlsh:bookstore> CONSISTENCY QUORUM;
Consistency level set to QUORUM.
cqlsh:bookstore> SELECT * FROM bookstore.books
... WHERE name='salma_rf1';

  name      | category | year | title
-----+-----+-----+-----
 salma_rf1 |  fiction | 2026 | Book_RF1

(1 rows)
cqlsh:bookstore> 
```

QUORUM = SUCCESS

➤ Consistency level = ONE

```
(1 rows)
cqlsh:bookstore> CONSISTENCY ONE;
Consistency level set to ONE.
cqlsh:bookstore> SELECT * FROM bookstore.books
... WHERE name='salma_rf1';

name          | category | year | title
-----+-----+-----+-----
salma_rf1     | fiction  | 2026 | Book_RF1

(1 rows)
cqlsh:bookstore> exit
```

ONE = SUCCESS

2- Second experiment: Node 2 down

Unpause Node 1 and Pause Node 2:

```
cqlsh:bookstore> exit
salma-rabie@salma-rabie-VirtualBox:~/Lab3$ docker unpause cassandra-seed
cassandra-seed
salma-rabie@salma-rabie-VirtualBox:~/Lab3$ docker exec cassandra-seed nodetool status
Datacenter: se1
=====
Status=Up/Down
|/ State=Normal/Leaving/Joining/Moving
-- Address      Load          Tokens   Owns (effective)  Host ID                               Rack
UN  172.18.0.4    92.75 KiB     16       35.7%             f32e8542-89b7-4297-af23-d3ba61b3bcd6  rack1
UN  172.18.0.3    92.74 KiB     16       31.6%             0b1b1cbd-57b0-4fb8-9fd6-cd95bdec4e85  rack1
UN  172.18.0.2    144.34 KiB    16       32.7%             60ed1d18-dc66-4a5d-863e-35ddee56228e  rack1

salma-rabie@salma-rabie-VirtualBox:~/Lab3$ docker pause cassandra1
cassandra1
salma-rabie@salma-rabie-VirtualBox:~/Lab3$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        NAMES
PORTS
e6b021a1e325   cassandra:4.1 "docker-entrypoint.s..." 33 minutes ago Up 33 minutes cassandra2
7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9040->9042/tcp, [::]:9040->9042/tcp
ce2fa6d9ac00   cassandra:4.1 "docker-entrypoint.s..." 35 minutes ago Up 35 minutes (Paused) cassandra1
7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9041->9042/tcp, [::]:9041->9042/tcp
f22c62757821   cassandra:4.1 "docker-entrypoint.s..." 35 minutes ago Up 35 minutes
7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9042->9042/tcp, [::]:9042->9042/tcp cassandra-seed
salma-rabie@salma-rabie-VirtualBox:~/Lab3$
```

Connect to Node 3:

➤ Consistency level = QUORUM

```
salma-rabie@salma-rabie-VirtualBox:~/Lab3$ docker exec -it cassandra2 cqlsh

ATTENTION: All commands will be saved to history file: /root/.cassandra/cqlsh_history
This may include sensitive information such as passwords.
To disable history, use --disable-history or set 'disabled = true' in the [history] section of cqlshrc.
See https://cassandra.apache.org/doc/latest/tools/cqlsh.html for more information.

Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.1.0 | Cassandra 4.1.12 | CQL spec 3.4.6 | Native protocol v5]
Use HELP for help.
cqlsh> USE bookstore;
cqlsh:bookstore> CONSISTENCY QUORUM;
Consistency level set to QUORUM.
cqlsh:bookstore> SELECT * FROM bookstore.books
... WHERE name='salma_rf1';

name      | category | year | title
-----+-----+-----+-----
salma_rf1 | fiction  | 2026 | Book_RF1

(1 rows)
```

QUORUM = SUCCESS

➤ Consistency level = ONE

```
cqlsh:bookstore> CONSISTENCY ONE;
Consistency level set to ONE.
cqlsh:bookstore> SELECT * FROM bookstore.books
... WHERE name='salma_rf1';

name      | category | year | title
-----+-----+-----+-----
salma_rf1 | fiction  | 2026 | Book_RF1

(1 rows)
cqlsh:bookstore> █
```

ONE = SUCCESS

3- Third experiment: Node 3 down

Unpause Node 2 and Pause Node 3:

```
salma-rabie@salma-rabie-VirtualBox:~/Lab3$ docker unpause cassandra1
cassandra1
salma-rabie@salma-rabie-VirtualBox:~/Lab3$ docker exec cassandra-seed nodetool status
Datacenter: se1
=====
Status=Up/Down
-- State=Normal/Leaving/Joining/Moving
-- Address      Load          Tokens   Owns (effective)  Host ID                               Rack
UN 172.18.0.4    92.75 KiB     16       35.7%             f32e8542-89b7-4297-af23-d3ba61b3bcd6 rack1
UN 172.18.0.3    92.74 KiB     16       31.6%             0b1b1cbd-57b0-4fb8-9fd6-cd95bdec4e85 rack1
UN 172.18.0.2    144.34 KiB    16       32.7%             60ed1d18-dc66-4a5d-863e-35ddee56228e rack1

salma-rabie@salma-rabie-VirtualBox:~/Lab3$ docker pause cassandra2
cassandra2
salma-rabie@salma-rabie-VirtualBox:~/Lab3$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        NAMES
e6b021a1e325   cassandra:4.1 "docker-entrypoint.s..." 48 minutes ago Up 48 minutes (Paused) cassandra2
ce2fa6d9ac00   cassandra:4.1 "docker-entrypoint.s..." 50 minutes ago Up 50 minutes cassandra1
f22c62757821   cassandra:4.1 "docker-entrypoint.s..." 50 minutes ago Up 50 minutes cassandra-seed
```

Connect to Node 1:

➤ Consistency level = QUORUM

```
salma-rabie@salma-rabie-VirtualBox:~/Lab3$ docker exec -it cassandra-seed cqlsh

ATTENTION: All commands will be saved to history file: /root/.cassandra/cqlsh_history
This may include sensitive information such as passwords.
To disable history, use --disable-history or set 'disabled = true' in the [history] section of cqlshrc.
See https://cassandra.apache.org/doc/latest/tools/cqlsh.html for more information.

Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.1.0 | Cassandra 4.1.12 | CQL spec 3.4.6 | Native protocol v5]
Use HELP for help.
cqlsh> USE bookstore;
cqlsh:bookstore> CONSISTENCY QUORUM;
Consistency level set to QUORUM.
cqlsh:bookstore> SELECT * FROM bookstore.books
... WHERE name='salma_rf1';
NoHostAvailable: ('Unable to complete the operation against any hosts', {<Host: 127.0.0.1:9042 se1>: Unavailable('Error from server: code=1000 [Unavailable exception] message="Cannot achieve consistency level QUORUM" info={\''consistency\'': \'QUORUM\'', \'required_replicas\'': 1, \'alive_replicas\'': 0}})'))
```

QUORUM = FAILURE

➤ Consistency level = ONE

```
cqlsh:bookstore> CONSISTENCY ONE;  
Consistency level set to ONE.  
cqlsh:bookstore> SELECT * FROM bookstore.books  
... WHERE name='salma_rf1';  
NoHostAvailable: ('Unable to complete the operation against any hosts', {<Host: 127.0.0.1:9042 se  
l>: Unavailable('Error from server: code=1000 [Unavailable exception] message="Cannot achieve con  
sistency level ONE" info={\'consistency\': \'ONE\', \'required_replicas\': 1, \'alive_replicas\':  
0}'))})  
cqlsh:bookstore>
```

ONE = FAILURE

Nodes Down	RF = 1, QUORUM	RF = 1, ONE
Node 1	SUCCESS	SUCCESS
Node 2	SUCCESS	SUCCESS
Node 3	FAILURE	FAILURE

CONCLUSION:

Because RF=1 stores only one replica, when that particular replica which is in Node 3 is on the failed node, lowering the consistency level from QUORUM to ONE cannot help. There are zero live replicas available.